

Introduction to Engineering Data Analytics with Python

Case Study

Can Çağınca | TU Berlin | QW

Introduction Data Analysis for Engineers with Python

The Final Project

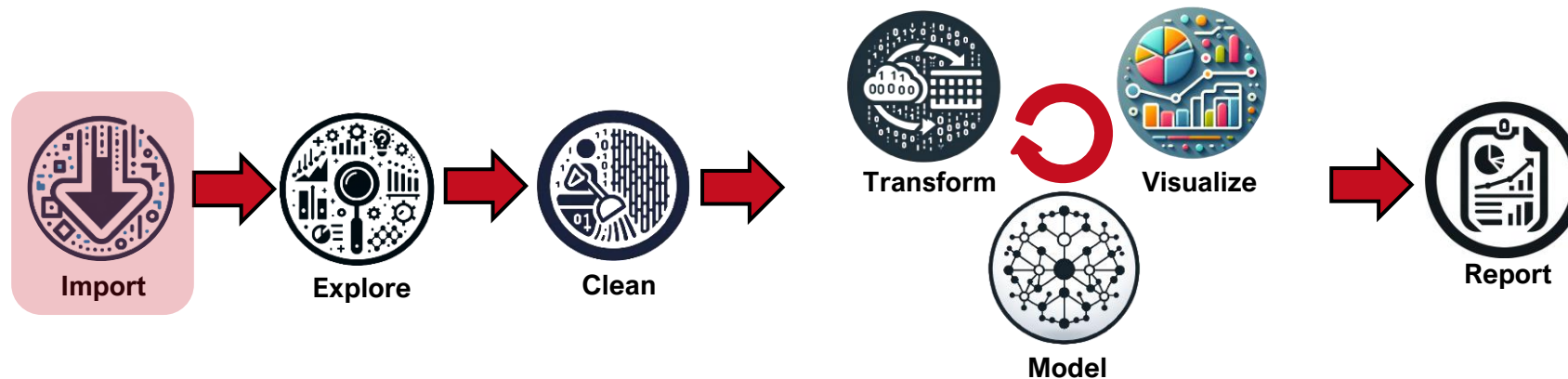
Import

Resource Management:

- Be mindful of RAM limits and avoid importing unnecessary data to optimize performance.
- Plan your data analysis steps to make the most efficient use of available resources.

Intelligent Data Importing:

- Use intelligent importing functions and scripts to handle data efficiently.
- Automate repetitive tasks and ensure your code is both efficient and readable.



Introduction Data Analysis for Engineers with Python

The Final Project

Preparing Data for Analysis

Transfer to a Suitable, Clean Data Format:

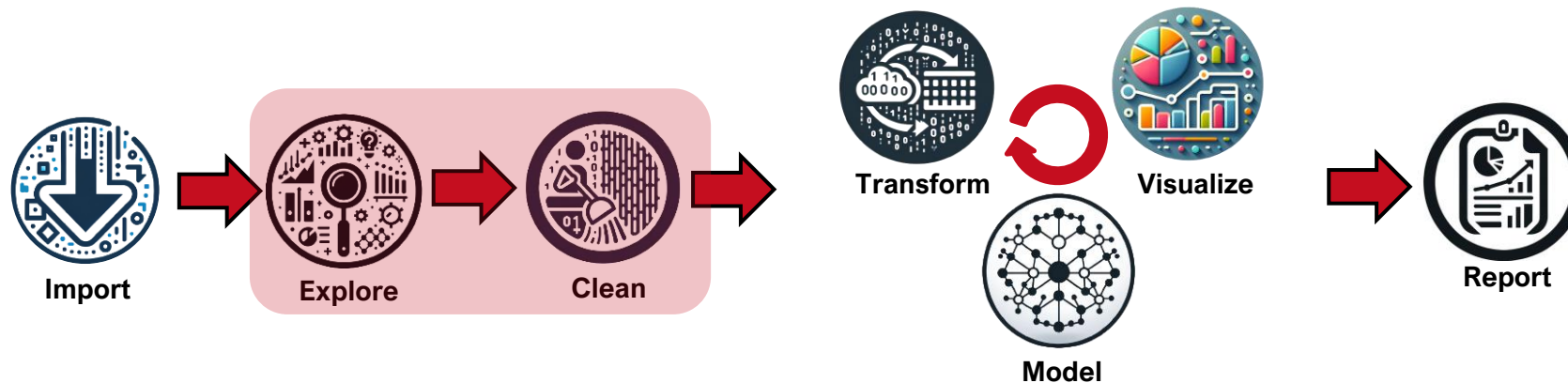
- **Convert Data:** Ensure your data is in a consistent format that your analysis tools can easily process.
- **Clean Data:** Remove any irrelevant or redundant information that may clutter the dataset.
- **Standardize:** Ensure all data entries follow the same standards, such as date formats, units of measurement, and text case.

Add Necessary Information:

- **Enhance Data:** Include any additional columns or metadata that will be crucial for your analysis.
- **Contextualize:** Add information that provides context to the data, such as geographic locations, time zones, or categorical labels.
- **Annotations:** Use annotations or comments to explain the origin or meaning of specific data points if necessary.

Check for Incorrect Information:

- **Identify Errors:** Look for anomalies or outliers that might indicate incorrect data, such as negative values for age or impossible dates.
- **Consistency Checks:** Verify that the data is consistent across different datasets if you're merging multiple sources.
- **Data Validation:** Use automated tools and scripts to validate the data, checking for missing values, duplicates, and logical inconsistencies.



Introduction Data Analysis for Engineers with Python

The Final Project

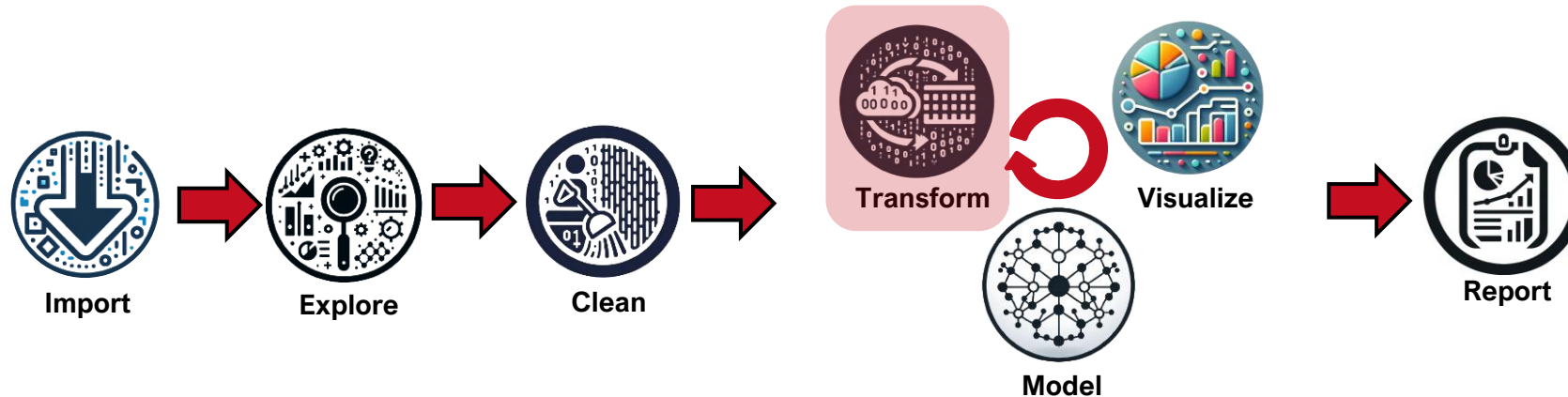
Transforming Data for Analysis

Reduce Data Size Sensibly:

- **Filter Unnecessary Data:** Remove columns and rows that are not relevant to your analysis to make the dataset more manageable.
- **Aggregate Data:** Summarize data by aggregating it, such as calculating averages or totals, to reduce the size without losing essential information.
- **Sample Data:** If the dataset is too large, consider using a representative sample that captures the key characteristics of the data.

Restructure Data in a Targeted Manner:

- **Normalize Data:** Convert data into a standard format to ensure consistency and comparability across different datasets.
- **Pivot Tables:** Use pivot tables to reorganize and summarize data, making it easier to analyze specific aspects.
- **Create New Features:** Derive new features or variables that can provide additional insights, such as combining date and time into a single timestamp or creating categorical variables from numerical ones.



Introduction Data Analysis for Engineers with Python

The Final Project

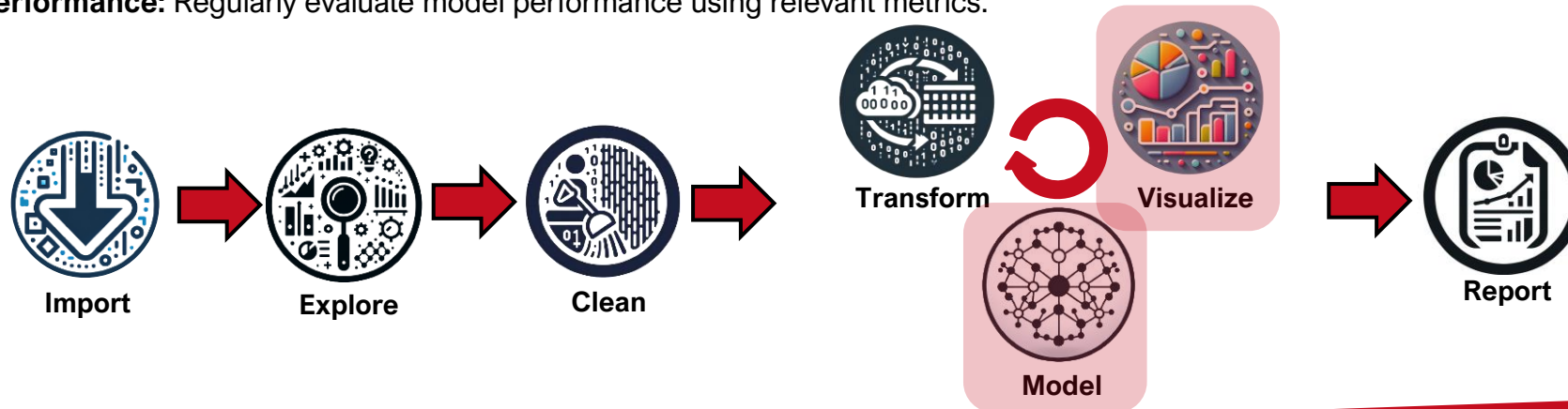
Visualizing and Modelling

Visualize and Understand Data:

- **Explore Data Visually:** Use proper visualization techniques such as histograms, scatter plots, and box plots to explore the data and identify patterns, trends, and outliers.
- **Interactive Visualizations:** Leverage interactive visualization tools like Plotly to gain deeper insights and allow for dynamic data exploration.
- **Create Clear Visualizations:** Use charts, graphs, and other visual aids to clearly present processes and your findings. Ensure your visualizations are easy to understand and interpret.
- **Tailor Visuals to the Audience:** Consider the knowledge level and interests of your audience. Use appropriate labels, legends, and annotations to make the visuals informative and engaging.
- **Automatization:** You can use defined functions for the frequently used plotting functions.
- **Documentation :** Document procedures and results in a customer-specific manner

Derive Target-Oriented Models:

- **Select Appropriate Models:** Based on your analysis goals, choose suitable models that best capture the underlying patterns and relationships.
- **Interpret Results:** Understand and interpret the model outputs to derive actionable insights and make informed decisions.
- **Evaluate Performance:** Regularly evaluate model performance using relevant metrics.



Introduction Data Analysis for Engineers with Python

The Final Project

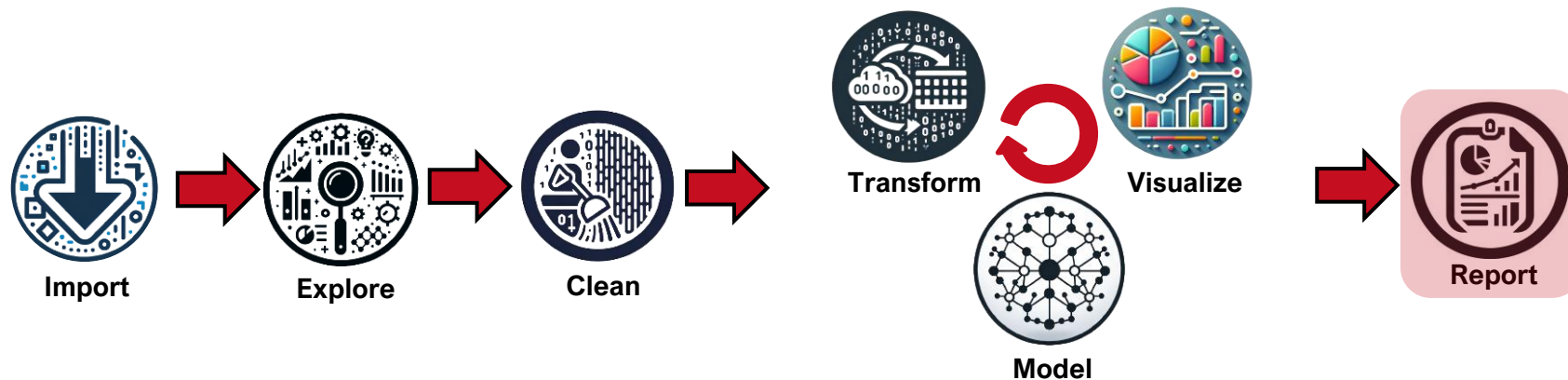
Reporting

Presentation of Findings:

1. **Duration** : Prepare a 10-minute presentation to showcase your findings from the data analysis.
2. **Content** : Focus on the key insights, visualizations, and conclusions derived from your analysis.
3. **Clarity** : Ensure your presentation is clear, concise, interactive and well-organized.

Question and Answer Session:

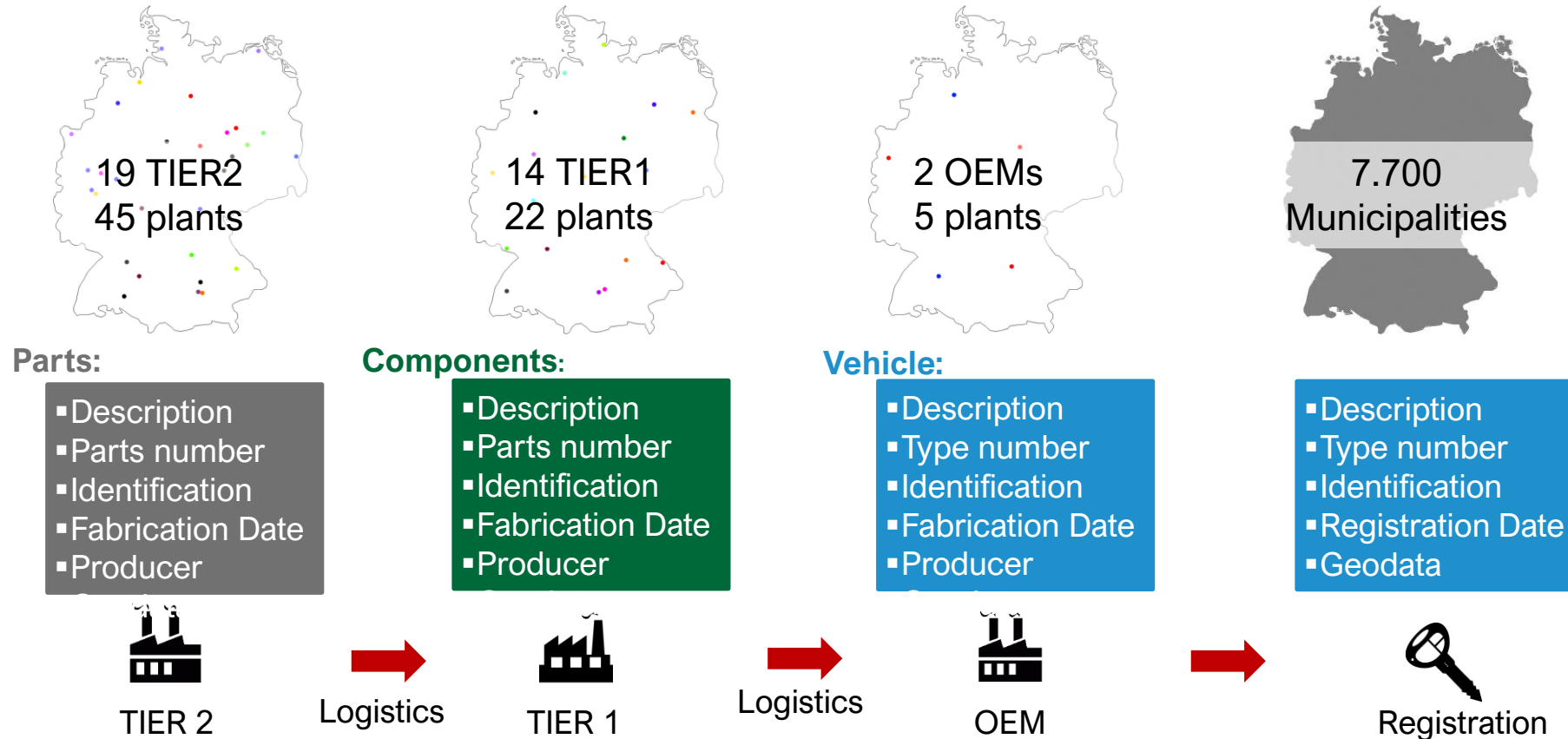
1. **Duration** : Allocate 10 minutes for a Q&A session.
2. **Topics** : Be prepared to answer questions regarding your data processing methods and the general tasks you undertook.
3. **Preparation** : Anticipate possible questions and practice your responses to ensure confidence during the session.



Introduction Data Analysis for Engineers with Python

The Final Project : Case Study

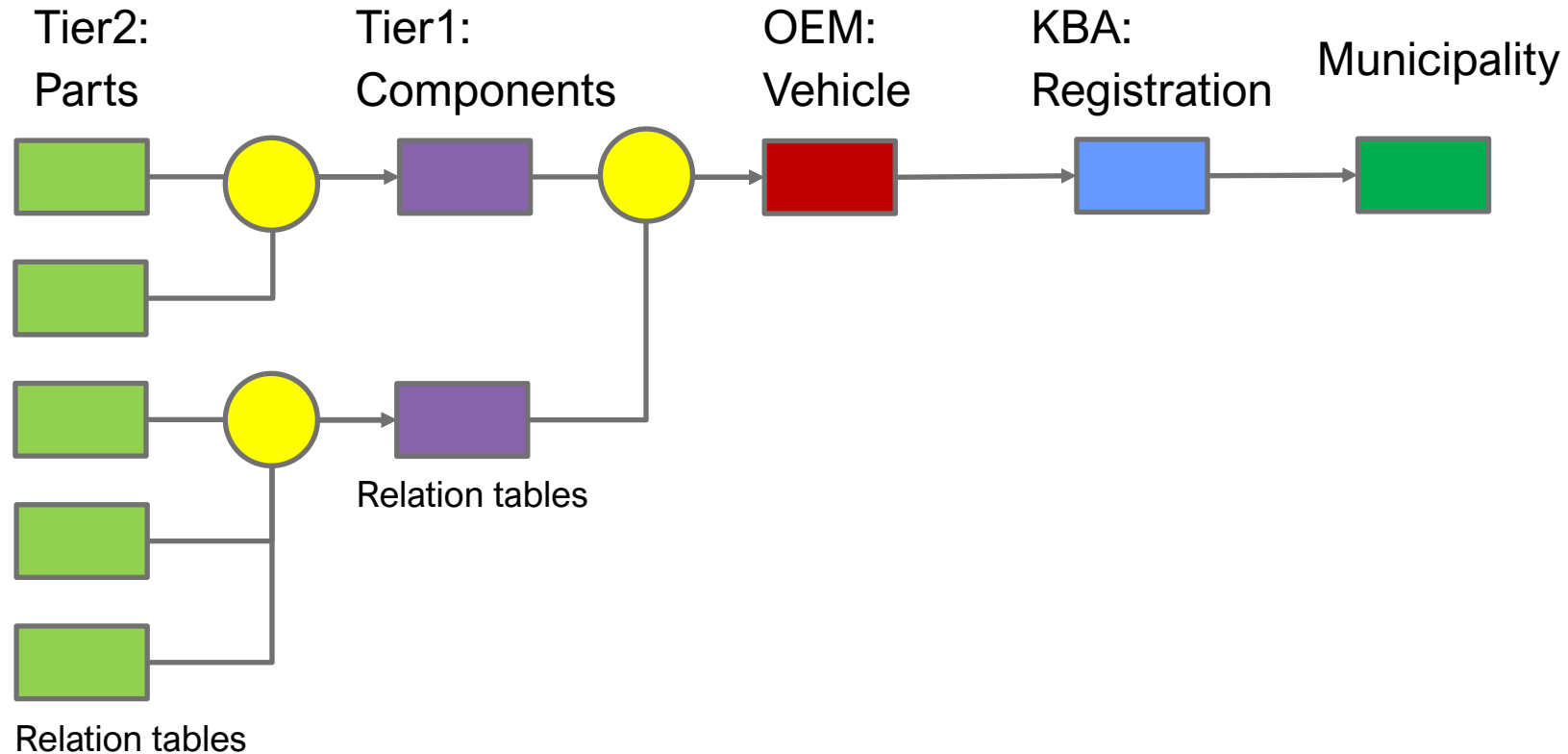
Dataset



Introduction Data Analysis for Engineers with Python

The Final Project : Case Study

Data Structure



Introduction Data Analysis for Engineers with Python

The Final Project : Case Study

Example for Parts Dataset

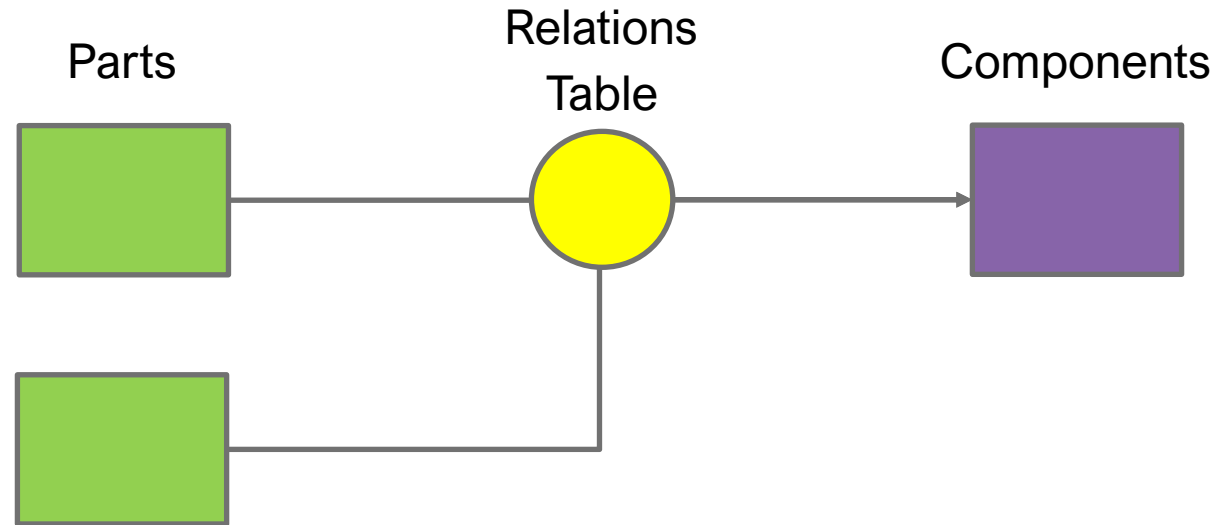
	X1	IDNummer	Produktionsdatum	Herstellernummer	Werksnummer	Fehlerhaft
1	768	2-201-2011-1	2008-11-07	201	2011	0
2	769	2-201-2011-2	2008-11-07	201	2011	0
3	770	2-201-2011-3	2008-11-07	201	2011	0
4	771	2-201-2011-4	2008-11-07	201	2011	0
5	772	2-201-2011-5	2008-11-07	201	2011	0
6	773	2-201-2011-6	2008-11-07	201	2011	0
7	774	2-201-2011-7	2008-11-07	201	2011	0
8	775	2-201-2011-8	2008-11-07	201	2011	0
9	776	2-201-2011-9	2008-11-07	201	2011	0
10	777	2-201-2011-10	2008-11-07	201	2011	0
11	778	2-201-2011-11	2008-11-07	201	2011	0
12	779	2-201-2011-12	2008-11-07	201	2011	0

Introduction Data Analysis for Engineers with Python

The Final Project : Case Study

Construction of Data Structure

Relation tables contain the ID numbers of individual installed parts along with the ID number of the system in which they are installed.

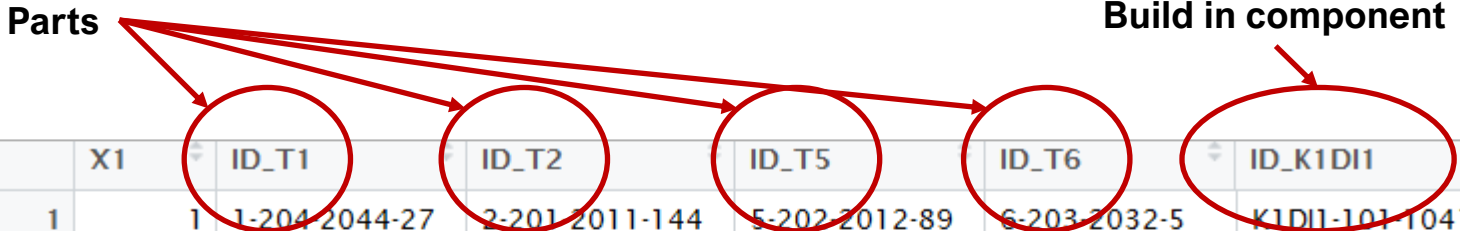


Introduction Data Analysis for Engineers with Python

The Final Project : Case Study

Example for Relation Table

Parts **Build in component**



	X1	ID_T1	ID_T2	ID_T5	ID_T6	ID_K1DI1
1	1	1-204-2044-27	2-201-2011-144	5-202-2012-89	6-203-2032-5	K1DI1-101-1041-1
2	2	1-202-2021-32	2-202-2022-577	5-202-2012-155	6-203-2032-79	K1DI1-101-1041-2
3	3	1-201-2011-238	2-202-2022-514	5-202-2012-199	6-203-2032-136	K1DI1-101-1041-3
4	4	1-202-2021-297	2-202-2022-626	5-201-2012-57	6-203-2032-147	K1DI1-101-1041-4
5	5	1-204-2044-73	2-201-2011-209	5-201-2012-157	6-204-2044-33	K1DI1-101-1041-5
6	6	1-201-2011-120	2-202-2022-453	5-201-2012-162	6-204-2044-58	K1DI1-101-1041-6
7	7	1-202-2021-134	2-202-2022-153	5-202-2012-3	6-204-2044-59	K1DI1-101-1041-7
8	8	1-201-2011-82	2-202-2022-693	5-202-2012-8	6-204-2044-110	K1DI1-101-1041-8
9	9	1-202-2021-176	2-202-2022-308	5-202-2012-13	6-203-2032-4	K1DI1-101-1041-9
10	10	1-201-2011-76	2-202-2022-219	5-202-2012-14	6-203-2032-8	K1DI1-101-1041-10
11	11	1-204-2044-118	2-202-2022-105	5-202-2012-16	6-203-2032-9	K1DI1-101-1041-11

Introduction Data Analysis for Engineers with Python

The Final Project

General Information

Accessing Tasks and Datasets:

- Download tasks and datasets from TUBcloud.
- Tasks will be available from 12.07.2022.

Submission Format:

- Submit your work in a .zip folder named **<Groupnumber>_SoSe2024_IDA_CaseStudy**.
- Upload your submission to the designated folder on TUBcloud.

Upload Details:

- There will be no confirmation message for a successful upload.
- The upload link expires on **01.09.2022** at 23:59, late submissions are not possible.
- Make sure your app runs on any computer. Avoid working in local repositories or working directories to ensure portability.
- **Document**, write down and justify every step of data processing with comments. This ensures transparency and helps others understand your methodology.

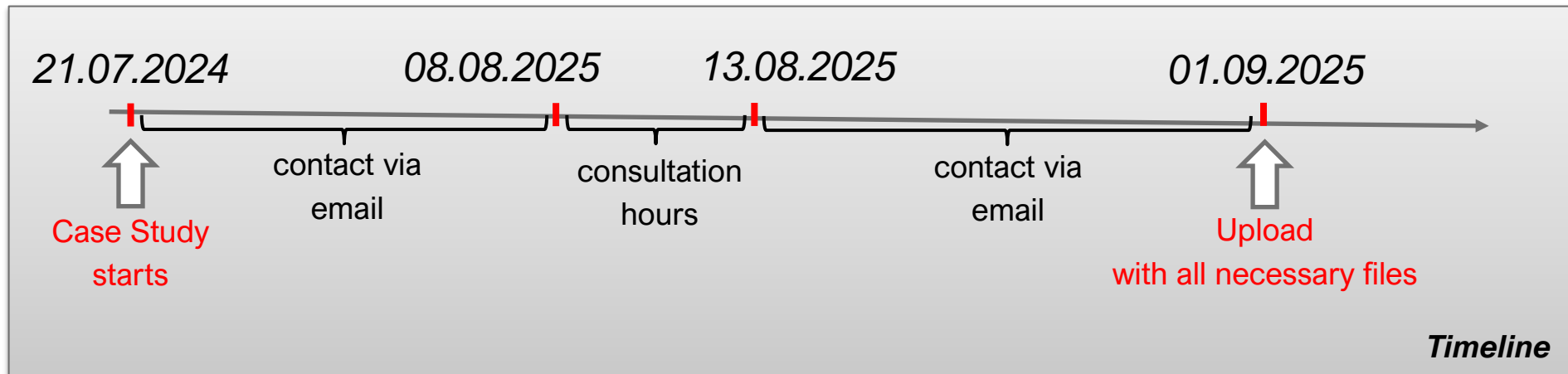
Introduction Data Analysis for Engineers with Python

The Final Project

Timeline

Support and Consultation:

- If you have problems or questions, email, attend the **consultation hours** or request an appointment.
- Regularly check ISIS announcements for updates and important information.



Introduction Data Analysis for Engineers with Python

The Final Project

Group Discussion and Task Understanding :

- Collaborate with your group to thoroughly discuss and understand the project task.
- If needed, arrange a consultation appointment with your instructor for further clarification.

Milestones and Deadlines:

- Set clear milestones, sub-goals, and deadlines to keep the project on track.
- Regularly review progress and adjust plans as necessary to meet these targets.

Strategic Planning :

- Determine the most favorable approach for tackling the project strategically.
- Consider different methods and choose the one that best suits your group's strengths and the project's requirements.

Resource Planning :

- Plan and allocate the necessary hardware and software resources for the project.
- Ensure all group members have access to the tools they need.

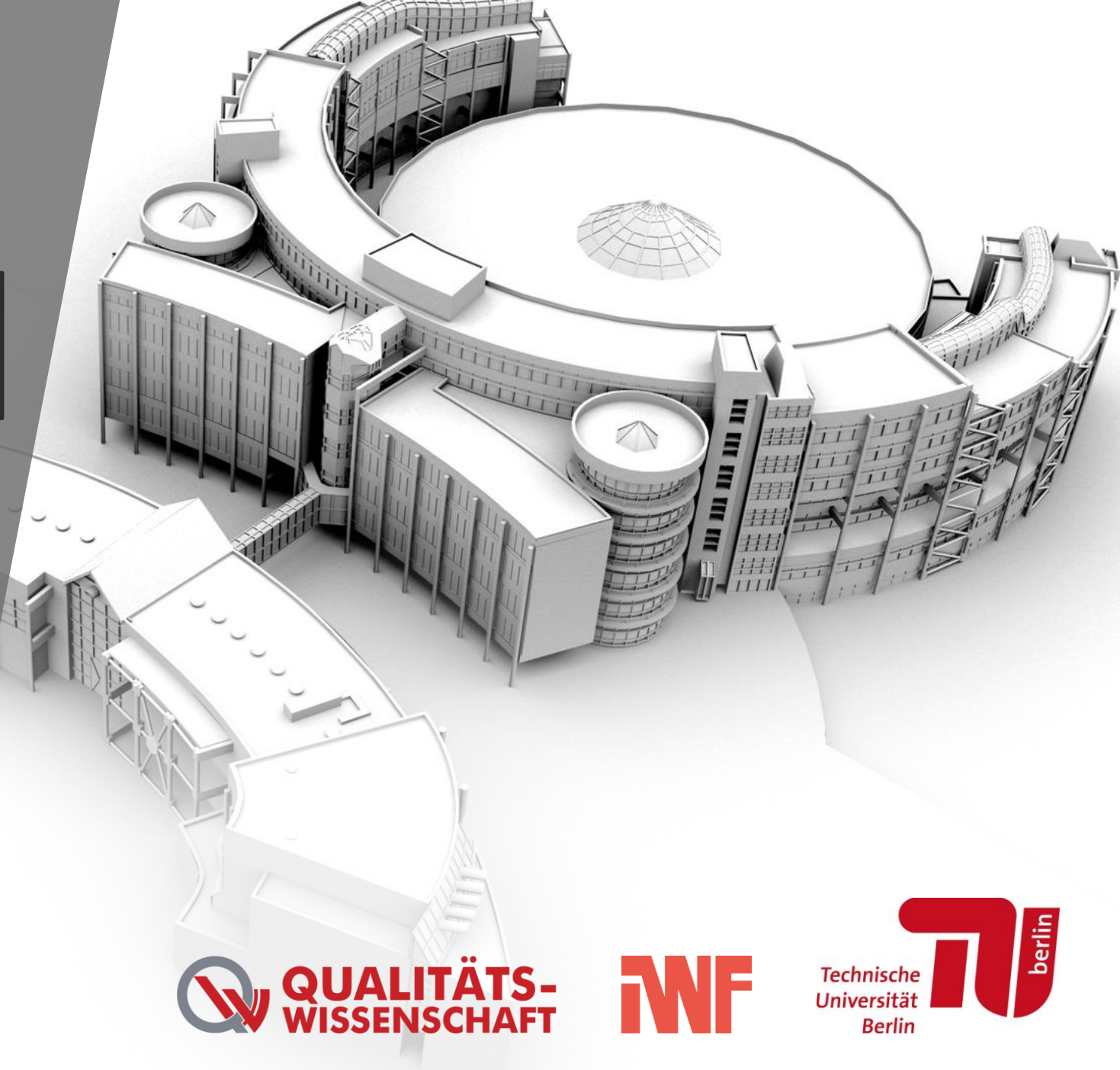
Work Distribution :

- Distribute work packages among group members based on their competencies and the effort required.
- Ensure that each member knows their responsibilities and deadlines.

Effective Communication :

- Maintain clear and consistent communication within the group.
- Schedule regular meetings to discuss progress, address issues, and make adjustments.

Questions ?



Thank you!

Technische Universität Berlin
Institut für Werkzeugmaschinen &
Fabrikbetrieb
Fachgebiet Qualitätswissenschaft
Pascalstr. 8-9 – 10587 Berlin

